

Week 3 — Narration & Scene Understanding

Goal: Add VLM narration, structured outputs, and action understanding.

Day	Key Tasks
Day 1	Integrate Claude/GPT-4V, Frame encoding
Day 2	Structured JSON prompting, Scene dict generation
Day 3	Hallucination reduction, Grounding prompts
Day 4	YOLOv8-Pose integration, Human action recognition
Day 5	Action vocabulary mapping, Standing/walking/sitting
Day 6	Natural language narration, Accessibility-style output
Day 7	Evaluation of narration quality, Human readability testing

Expected Deliverables

- Real-time scene narration
- Structured JSON outputs
- Action-aware descriptions
- Reduced hallucinations

This week contributes directly toward building a complete real-time scene understanding research system capable of object detection, depth estimation, segmentation, narration, and intelligent perception.